

Practical no 7

- Name : Durga Solanke
- Branch : AI&DS
- Roll No : SY B 47
- Batch : SY B3

1. Difference between Binomial, Poisson, and Normal Distributions

The Binomial distribution is used when we perform an experiment a fixed number of times and each trial has only two possible outcomes, like success or failure. For example, tossing a coin 10 times and counting how many times we get heads.

The Poisson distribution is used when we want to count how many times an event happens in a fixed interval of time or space. These events are usually rare, such as the number of accidents at a traffic signal in a day.

The Normal distribution, also known as the bell-shaped curve, is used for continuous data like height, weight, or marks. It shows how values are spread around an average.

In simple terms:

Binomial deals with fixed trials Poisson deals with event frequency Normal deals with continuous natural data

2. Real-World Significance of Poisson Distribution

The Poisson distribution plays a very important role in real life because it helps us model situations where events happen randomly over time.

For example, it can be used to predict:

How many customers arrive at a bank in one hour
How many phone calls a call center receives in a minute
How many defects are found in a product batch

This distribution is useful because it helps businesses and systems prepare for uncertainty. For instance, a hospital can estimate how many emergency cases may arrive, or a company can manage server traffic more efficiently.

3. Central Limit Theorem (CLT) and its Relation to Normal Distribution

The Central Limit Theorem (CLT) is one of the most important concepts in statistics. It states that when we take a large number of samples from any distribution, the average of those samples will follow a Normal distribution, even if the original data is not normally distributed.

$$\bar{X} \sim N(\mu, \frac{\sigma}{\sqrt{n}})$$

σ

)

This is very powerful because it allows us to use Normal distribution techniques in many real-world problems, even when the data itself is irregular or skewed.

4. Why Normal Distribution is Widely Used

The Normal distribution is widely used because it naturally appears in many real-life situations. For example, the height of people in a population or marks obtained by students often follow a bell-shaped pattern.

Another reason is that it is mathematically simple and easy to work with. It only depends on two values: the mean (average) and the standard deviation (spread).

Also, due to the Central Limit Theorem, even non-normal data can be approximated using a Normal distribution when the sample size is large. Because of all these reasons, it is heavily used in statistics, data analysis, and machine learning.

5. Skewness in Distributions

Skewness refers to how symmetrical or asymmetrical a distribution is.

If a distribution is perfectly balanced, it has zero skewness, like the Normal distribution.

If the tail is longer on the right side, it is called positive skewness, which is commonly seen in the Poisson distribution.

If the tail is longer on the left side, it is called negative skewness.

The Binomial distribution can be either symmetrical or skewed depending on the probability of success.

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import binom, poisson, norm

# Set up a figure with subplots
fig, axs = plt.subplots(3, 3, figsize=(15, 12))
fig.suptitle("Analysis of Binomial, Poisson, and Normal Distributions")

# ----- BINOMIAL DISTRIBUTION -----
n, p = 10, 0.5
x_binom = np.arange(0, n+1)
pmf_binom = binom.pmf(x_binom, n, p)
cdf_binom = binom.cdf(x_binom, n, p)
samples_binom = binom.rvs(n, p, size=1000)

# PMF
axs[0, 0].bar(x_binom, pmf_binom, color='skyblue')
axs[0, 0].set_title("Binomial PMF (n=10, p=0.5)")
# CDF
axs[0, 1].step(x_binom, cdf_binom, where='mid', color='blue')
```

```

axs[0, 1].set_title("Binomial CDF")
# Histogram of Samples
axs[0, 2].hist(samples_binom, bins=np.arange(n+2)-0.5, density=True,
axs[0, 2].set_title("Binomial Samples Histogram")

# ----- POISSON DISTRIBUTION -----
mu = 4
x_poisson = np.arange(0, 15)
pmf_poisson = poisson.pmf(x_poisson, mu)
cdf_poisson = poisson.cdf(x_poisson, mu)
samples_poisson = poisson.rvs(mu, size=1000)

# PMF
axs[1, 0].bar(x_poisson, pmf_poisson, color='lightgreen')
axs[1, 0].set_title("Poisson PMF ( $\mu=4$ )")
# CDF
axs[1, 1].step(x_poisson, cdf_poisson, where='mid', color='green')
axs[1, 1].set_title("Poisson CDF")
# Histogram of Samples
axs[1, 2].hist(samples_poisson, bins=np.arange(16)-0.5, density=True,
axs[1, 2].set_title("Poisson Samples Histogram")

# ----- NORMAL DISTRIBUTION -----
mu_norm, sigma = 0, 1
x_norm = np.linspace(-4, 4, 1000)
pdf_norm = norm.pdf(x_norm, mu_norm, sigma)
cdf_norm = norm.cdf(x_norm, mu_norm, sigma)
samples_norm = norm.rvs(mu_norm, sigma, size=1000)

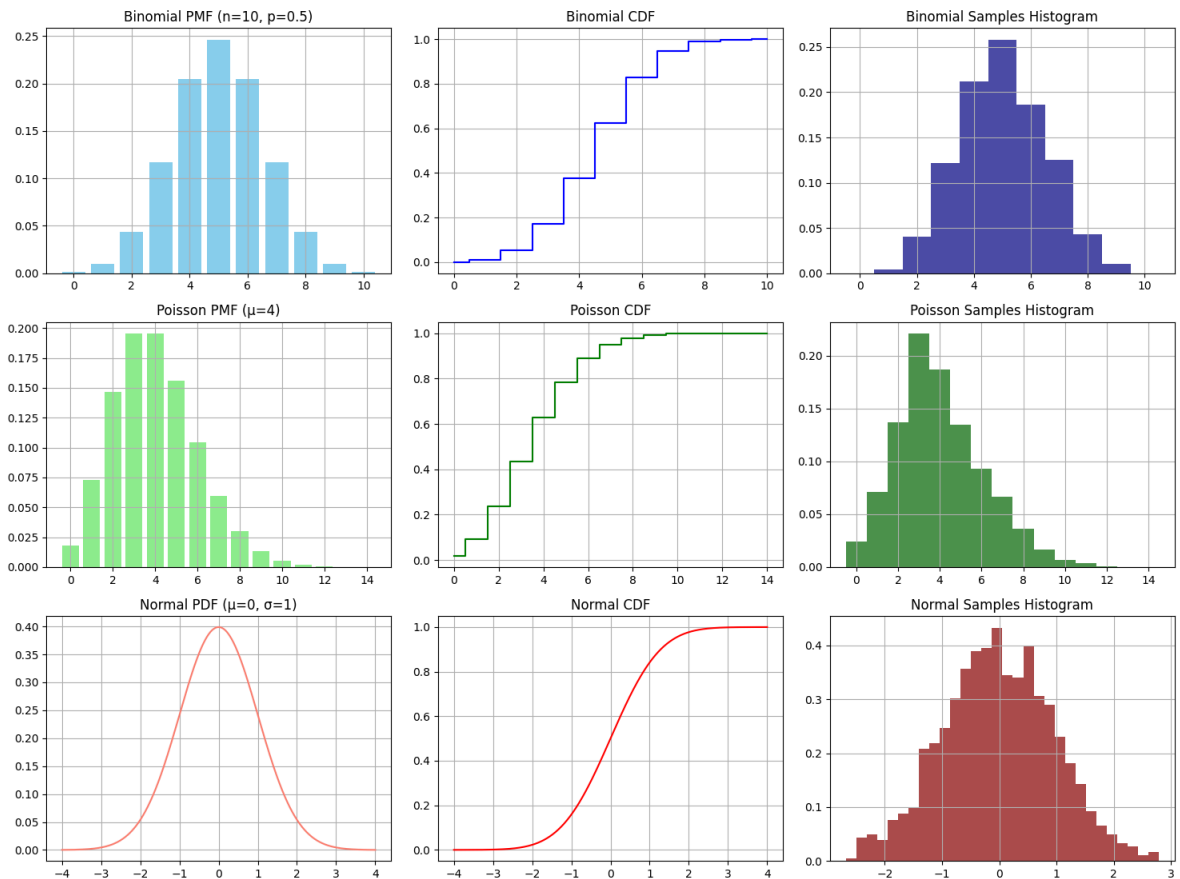
# PDF
axs[2, 0].plot(x_norm, pdf_norm, color='salmon')
axs[2, 0].set_title("Normal PDF ( $\mu=0$ ,  $\sigma=1$ )")
# CDF
axs[2, 1].plot(x_norm, cdf_norm, color='red')
axs[2, 1].set_title("Normal CDF")
# Histogram of Samples
axs[2, 2].hist(samples_norm, bins=30, density=True, color='darkred',
axs[2, 2].set_title("Normal Samples Histogram")

# Layout and show
for row in axs:
    for ax in row:
        ax.grid(True)
plt.tight_layout(rect=[0, 0, 1, 0.96])
plt.show()

# ----- BASIC STATISTICAL ANALYSIS -----
print("Binomial Distribution:")
print(f"Mean = {np.mean(samples_binom):.2f}, Variance = {np.var(samples_binom):.2f}")
print("\nPoisson Distribution:")
print(f"Mean = {np.mean(samples_poisson):.2f}, Variance = {np.var(samples_poisson):.2f}")
print("\Normal Distribution:")
print(f"Mean = {np.mean(samples_norm):.2f}, Standard Deviation = {np.std(samples_norm):.2f}")

```

Analysis of Binomial, Poisson, and Normal Distributions



Binomial Distribution:
Mean = 5.01, Variance = 2.36

Poisson Distribution:
Mean = 4.03, Variance = 4.41

Normal Distribution:
Mean = -0.02, Standard Deviation = 0.97

Conclusion In conclusion, Binomial, Poisson, and Normal distributions are essential tools in statistics. Each of them is used in different situations depending on the type of data. The Normal distribution is especially important because of its wide applicability and its connection to the Central Limit Theorem. Understanding these distributions helps us analyze data better and make informed decisions.

